

LEC32 - Power Series and Functions

Warm Up Problem

Which of the following statements are true?

1. $\sum_{n=1}^{\infty} 5^n x^n$ is both a power series and a geometric series **True**
2. A power series can converge for some values of x and diverge for other x **True**
3. $\sum_{n=0}^{\infty} \frac{n!}{(x-4)^n}$ is a power series centered at $x=4$ **False**

Applications of Power Series

1. Representing functions as power series and therefore...

- Approximating values easily (π , e , $\sqrt{5}$, $\ln(15)$, $\sin(1)$, etc.)
- Evaluating integrals
- Solving equations
- Computing limits
- Solving differential equations and other problems not seen in this course

2. Use power series to define new functions

- Examples include Bessel functions and Legendre polynomials
- Many applications in physics, astronomy, chemistry, statistics, engineering, and also topics within mathematics

Example

Suppose that $\sum_{n=0}^{\infty} c_n (x-2)^n$ converges when $x=4$ and diverges when $x=6$.

What can we say about convergence at other integers?

	$x=2$	$x=3$	$x=4$	$x=5$	$x=6$	$x=7$
converges	✓	✓	✓			
diverges						✓
not enough information				✓	✓	

Example

Find the radius interval of convergence of the power series $\sum_{n=1}^{\infty} \frac{(2x-3)^n}{n \cdot 5^n}$

⇒ Proceed with ratio test

$$\Rightarrow \lim_{n \rightarrow \infty} \left| \frac{(2x-3)^{n+1}}{(n+1) \cdot 5^{n+1}} \cdot \frac{n \cdot 5^n}{(2x-3)^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(2x-3) \cdot n}{5(n+1)} \right| = \frac{2x-3}{5}$$

$$\Rightarrow -1 < \frac{2x-3}{5} < 1$$

$$\Rightarrow -5 < 2x-3 < 5$$

$$\Rightarrow -2 < 2x < 8$$

$$\Rightarrow -1 < x < 4$$

⇒ We need to check $x=-1$, $x=4$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{(-5)^n}{n \cdot 5^n}, \text{ converges by AST}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{(5)^n}{n \cdot 5^n} = \sum_{n=1}^{\infty} \frac{1}{n}, \text{ which diverges}$$

⇒ Therefore, the series converges for $x \in [-1, 4)$

Outline for Chapter 6

Section 6.1

- Definitions
- For what values of x does a power series converge?
- Finding a power series for certain functions

Section 6.2

- Finding and manipulating power series (combining, multiplying, differentiating, integrating...)

Section 6.3

- A systematic way of finding the power series of a function (Taylor and Maclaurin series)
- Approximating functions and function values using series

Section 6.4

- More applications (will also see some along the way)

Convergence of a Power Series

Any power series $\sum_{n=0}^{\infty} c_n (x-a)^n$ satisfies exactly one of the following:

- i) Converges only at $x=a$ ($R=0$)
- ii) Converges for all x ($R=\infty$)
- iii) Converges if $|x-a| < R$ and diverges if $|x-a| > R$, for some $R > 0$
 - Could converge or diverge when $|x-a| = R$

- A power series centered at a always converges at $x=a$ (all terms are 0 except $n=0$)
- x further away from a means less likely to converge (terms are larger)
- Sometimes the tipping point is close to a (i and iii), sometimes no tipping point (ii)